

LAVANYA RATHI

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Education

National Institute of Technology, Surathkal, Karnataka

August 2023-May 2027

Bachelor of Technology in Information Technology

CGPA: 7.89

Minor in Management

Minor CGPA: 8.0

Relevant Coursework

- DSA
- Operating Systems
- OOPs
- Database Management

Projects

AlgoArena: Collaborative coding platform | *React.js, Node.js, Socket.IO*

- Built a real-time collaborative coding platform using React.js, Node.js, and Socket.IO, enabling multiple users to code simultaneously with synchronized updates.
- Integrated live multi-language code execution via the Piston API and enhanced the coding experience with features like syntax highlighting and auto-completion using CodeMirror.
- Implemented secure user authentication, room-based session management, and real-time chat functionality for seamless developer communication.

Command Line Interpreter | *C, Linux*

- Designed and implemented a Unix-like shell in C with support for command execution, argument parsing, pipelines, and built-in commands like cd, alias, and history.
- Built modular components for parsing, execution, signal handling, and persistent history using file I/O, hash tables, and linked lists.
- Integrated advanced features such as pipe()-based inter-process communication, signal handling (SIGINT/SIGTSTP), redirection, and aliasing, with a scalable architecture for future enhancements.

SkillBridge | *MongoDB, Express.js, React.js, Node.js*

- Developed a skill-sharing platform with role-based access for Students and Instructors, enabling content creation and course enrollment.
- Created user dashboards in React.js to display enrolled courses, posted content, and instructor-specific features.
- Added features for instructors to upload learning materials (e.g., PDFs) and for students to submit course reviews, with all data stored in MongoDB collections

Technical Skills

Languages: C++, HTML, CSS, JavaScript, SQL

Developer Tools: VS Code, Eclipse, Postman, Linux

Technologies/Frameworks: Linux, Git, GitHub, MongoDB

Soft Skills: Team collaboration, Agile development, Technical communication

Extracurricular

ISTE NITK

Sept 2024 – Present

Executive Member-CRYPT

- Collaborated with fellow members on innovative projects, demonstrating strong teamwork and technical acumen.
- Led a team of 5 to execute 3+ high-profile tech events showcasing leadership and management skills

DDFC NITK

Dec 2023–Present

Executive Member-Marketing SIG

- Engineered social media campaigns (Instagram/WhatsApp) for 5+ college tech events, increasing engagement by 40% through data-driven scheduling

Achievements

- Secured AIR 6135 in JEE-Mains 2023, ranking in the top 0.52% out of 1.1 million candidates.
- JEE-Advanced 2023 with AIR 18522, outperforming 98% of 180,000 candidates.
- 3rd place (consecutive) in DDFC's *Bharat Darshan* - outperformed 8+ teams in state-representation competition